

2 . OBJECT DETECTION

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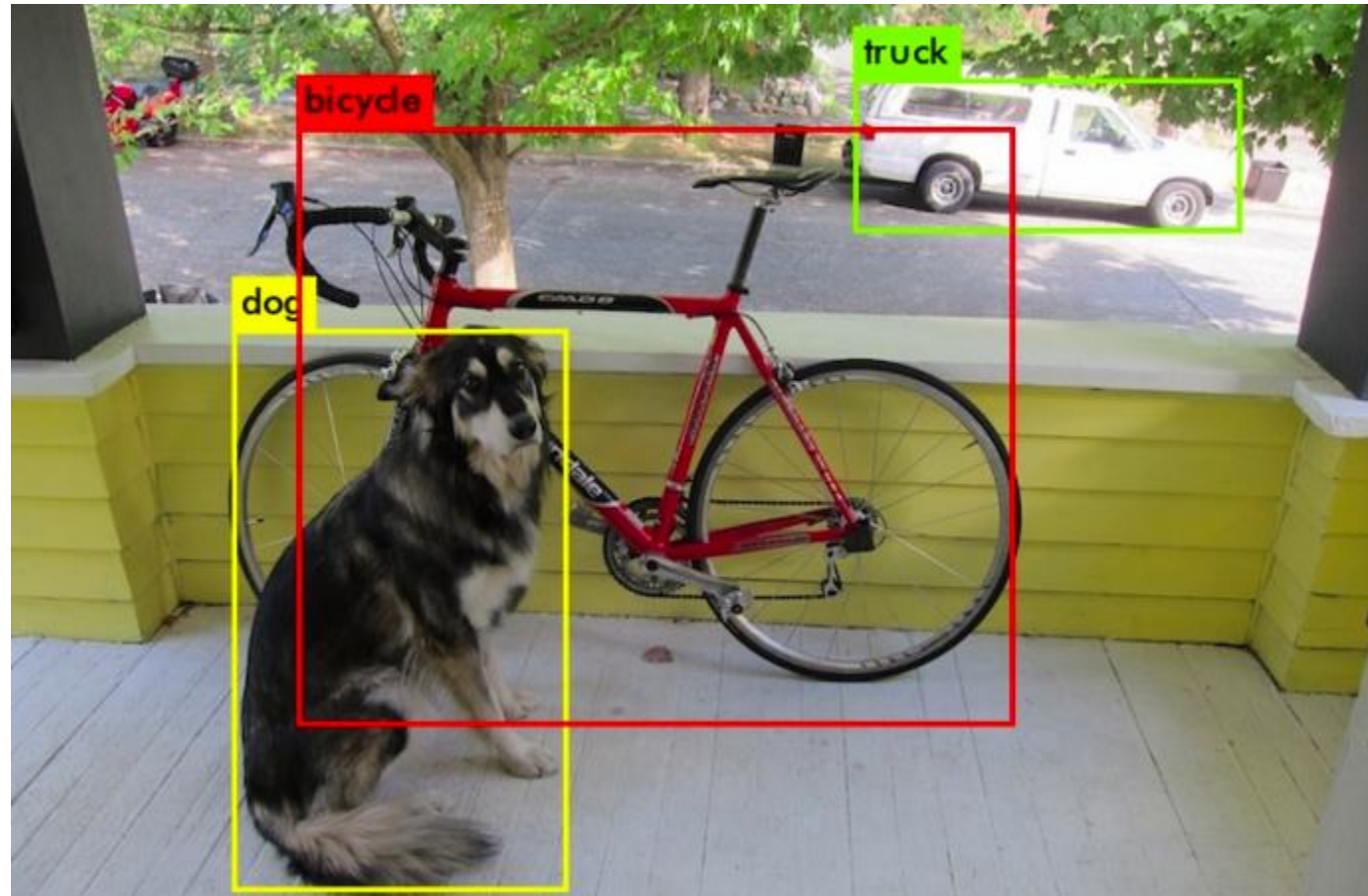
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1 . I N T R O D U C T I O N

DEFINITION OF THE PROBLEM

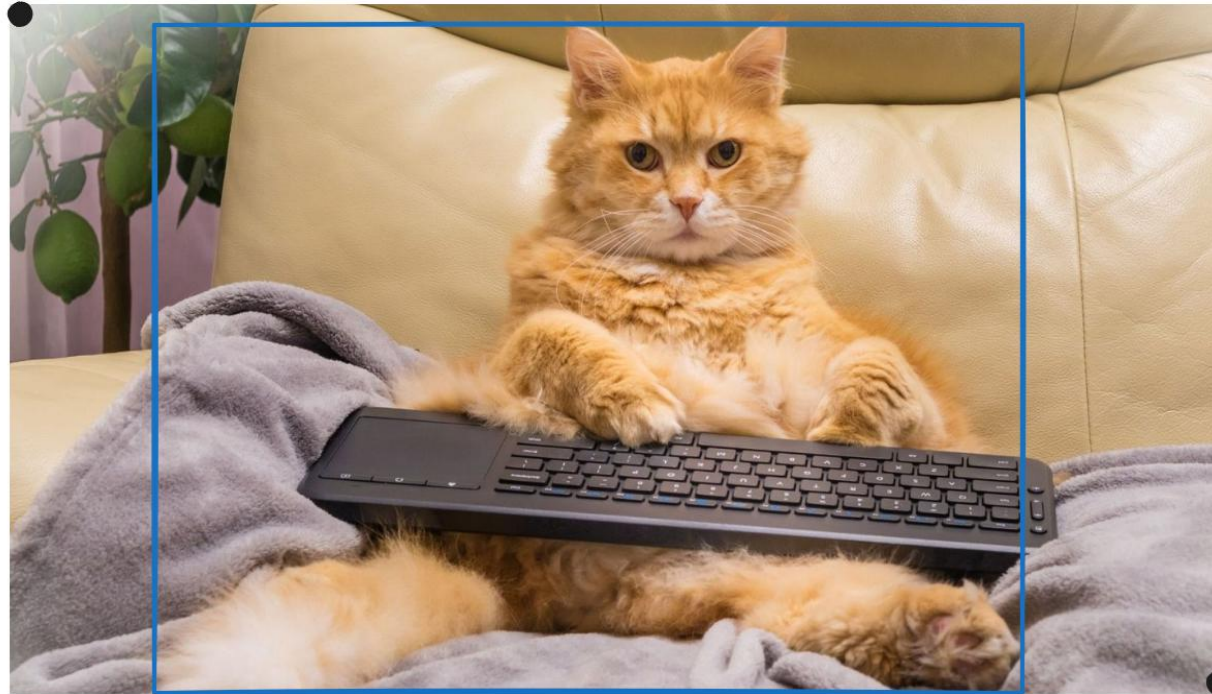


DEFINITION OF THE PROBLEM

Two ways of representing bounding boxes:

- **Upper-left and lower-right corners:** (10, 5, 85, 100)
- **Center, height and width:** (47, 52, 75, 95)

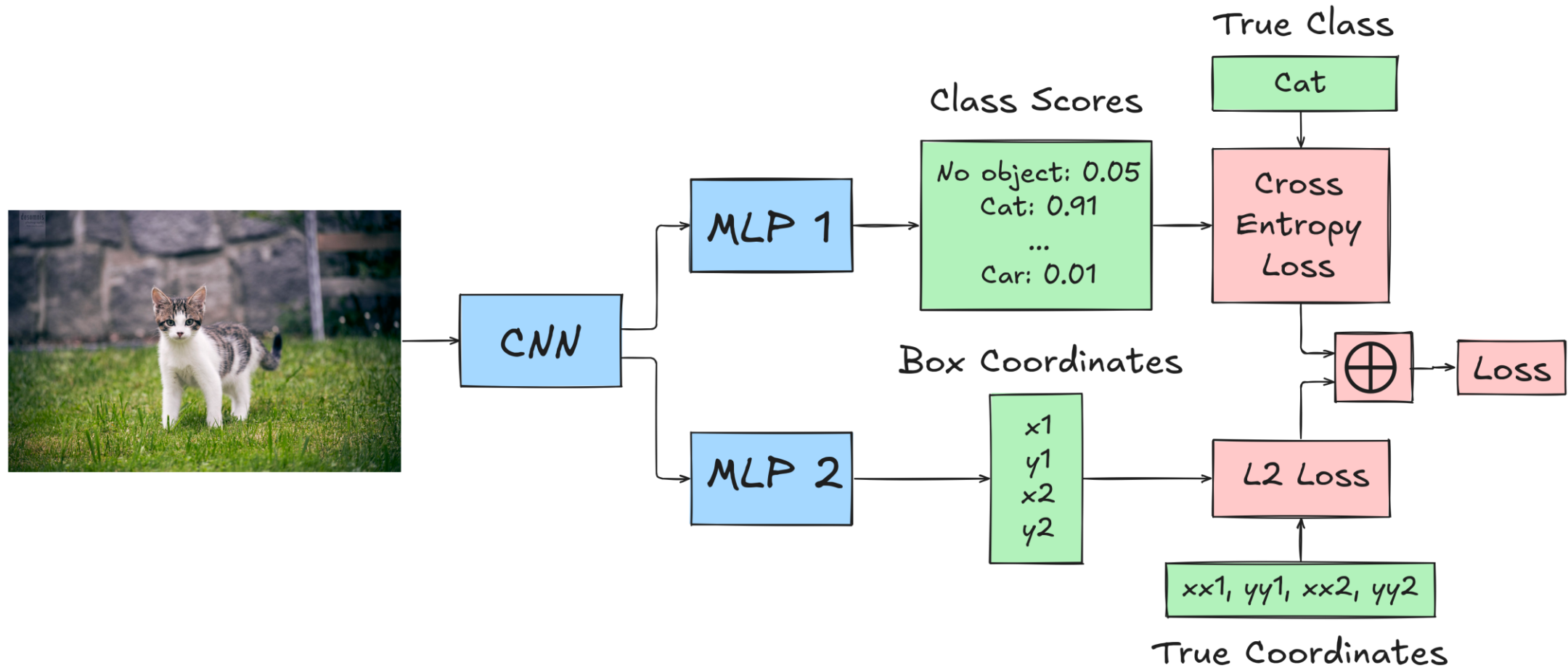
(0, 0)



(100, 100)



DETECTING A SINGLE OBJECT



DETECTING A SINGLE OBJECT

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \cdot 1_{\text{no_bg}} \cdot \mathcal{L}_{L_2}$$

Class of the box

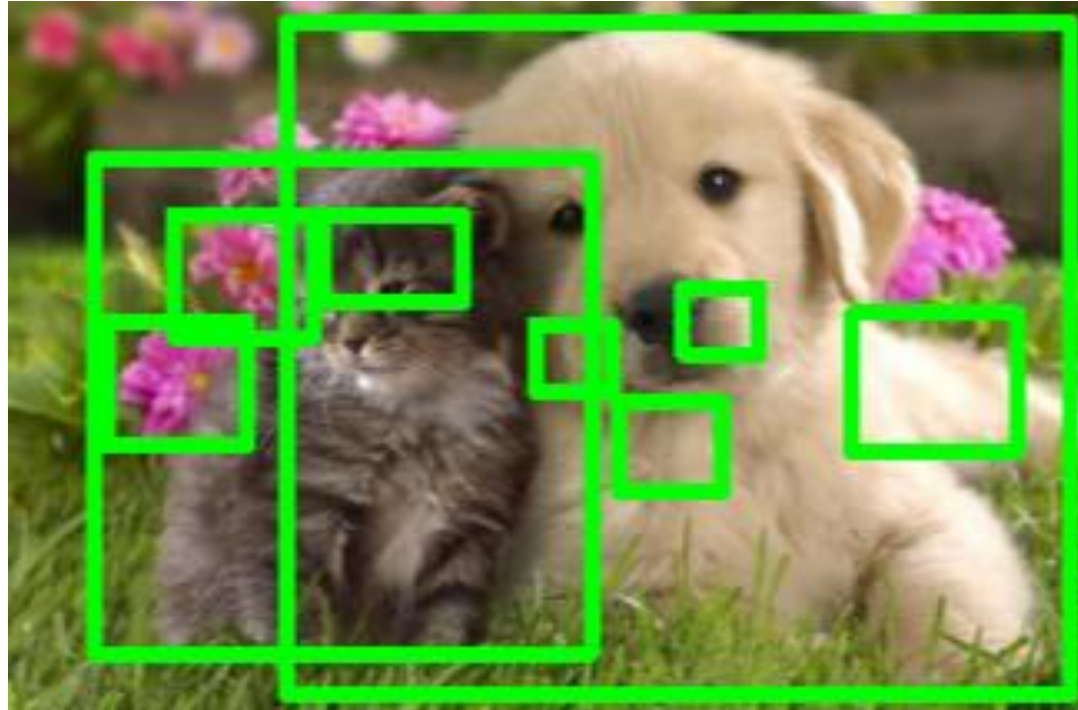
$$\mathcal{L}_{\text{CE}} = - \sum_{i=1}^n \log \left(\frac{\exp(\hat{y}_i)}{\sum_{j=1}^n \exp(y_j)} \right) y_i$$

Coordinates of the box:

$$\mathcal{L}_{L_2} = \frac{1}{4} \left((x_1 - \hat{x}_1)^2 + (y_1 - \hat{y}_1)^2 + (x_2 - \hat{x}_2)^2 + (y_2 - \hat{y}_2)^2 \right)$$



NAIVE APPROACH

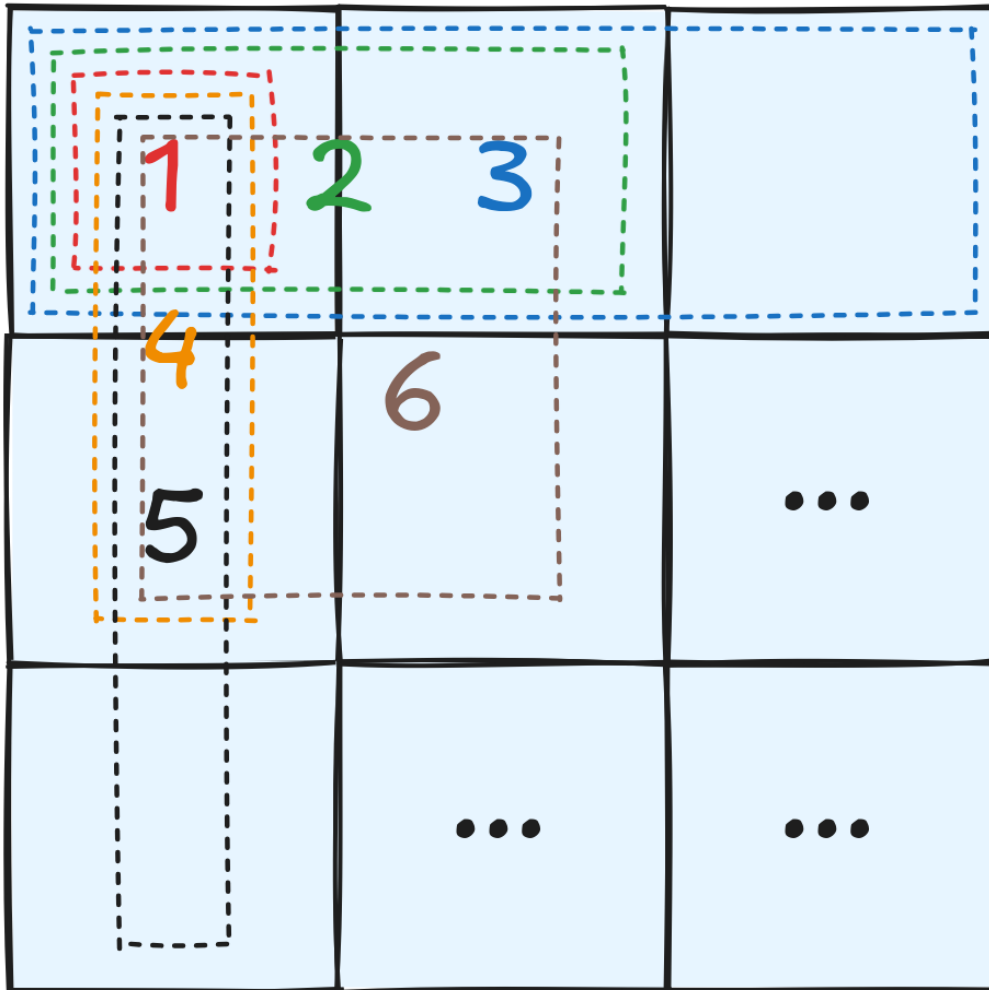


Problem: We can only detect one object

Solution: Create many crops and apply single object detection to each one of them!



NAIVE APPROACH



Problem: Too many possible bounding boxes!

$$\mathcal{O}(H^2W^2)$$



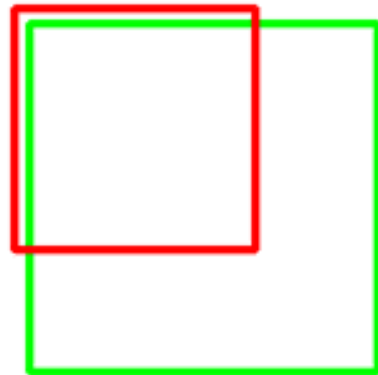
2 . P R E V I O U S C O N C E P T S

INTERSECTION OVER UNION

How can a "good" predicted bounding box be **quantified**?

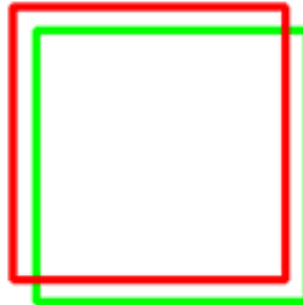
$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

IoU: 0.4034



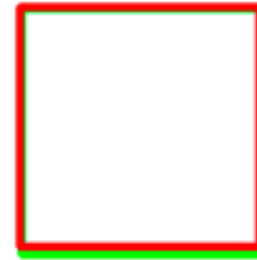
Poor

IoU: 0.7330



Good

IoU: 0.9264



Excellent

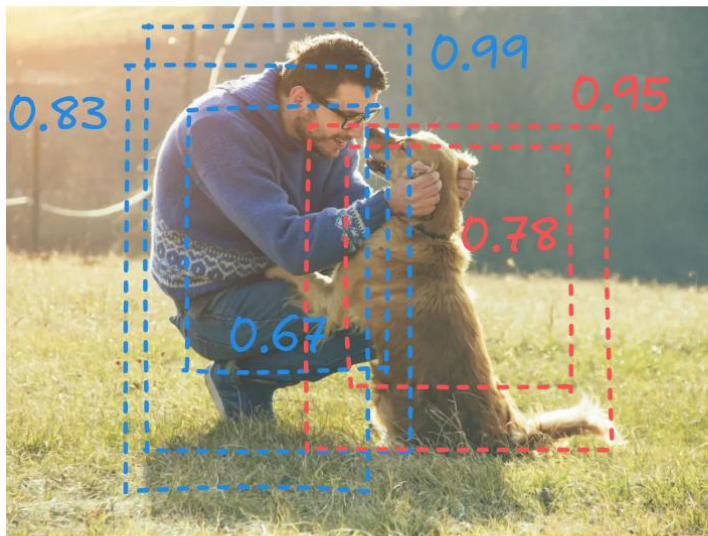


NON-MAXIMUM SUPPRESSION (NMS)

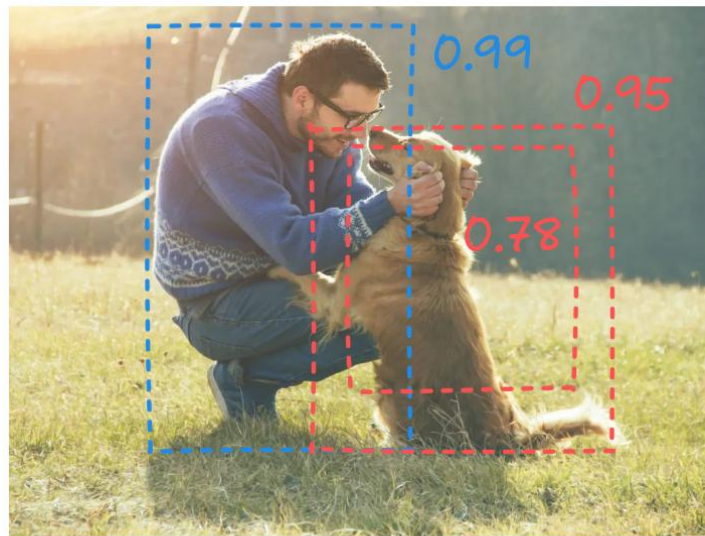
1. Order all predicted bounding boxes by confidence in a list L
2. Choose the one with most confidence that has not been checked and remove all similar bounding boxes in L
3. Repeat 2 until all bounding boxes have been checked
4. Return L



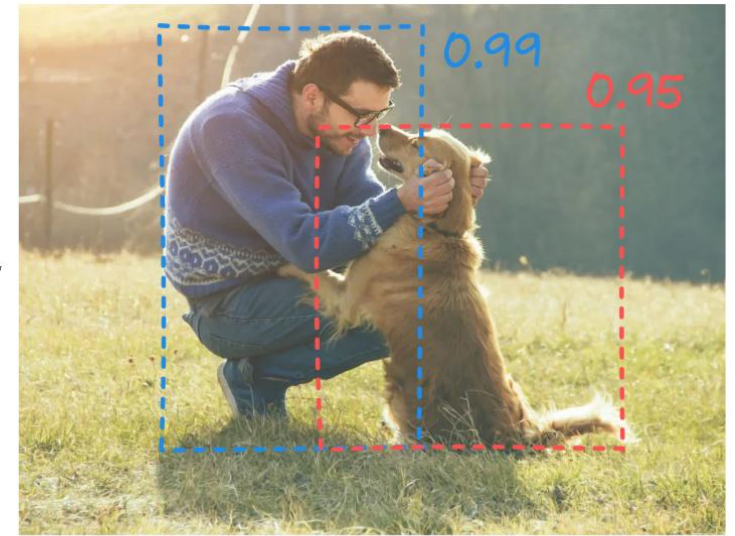
NON-MAXIMUM SUPPRESSION (NMS)



$L = [0.99, 0.95, 0.83, 0.78, 0.67]$



$L = [0.99, 0.95, 0.78]$



$L = [0.99, 0.95]$



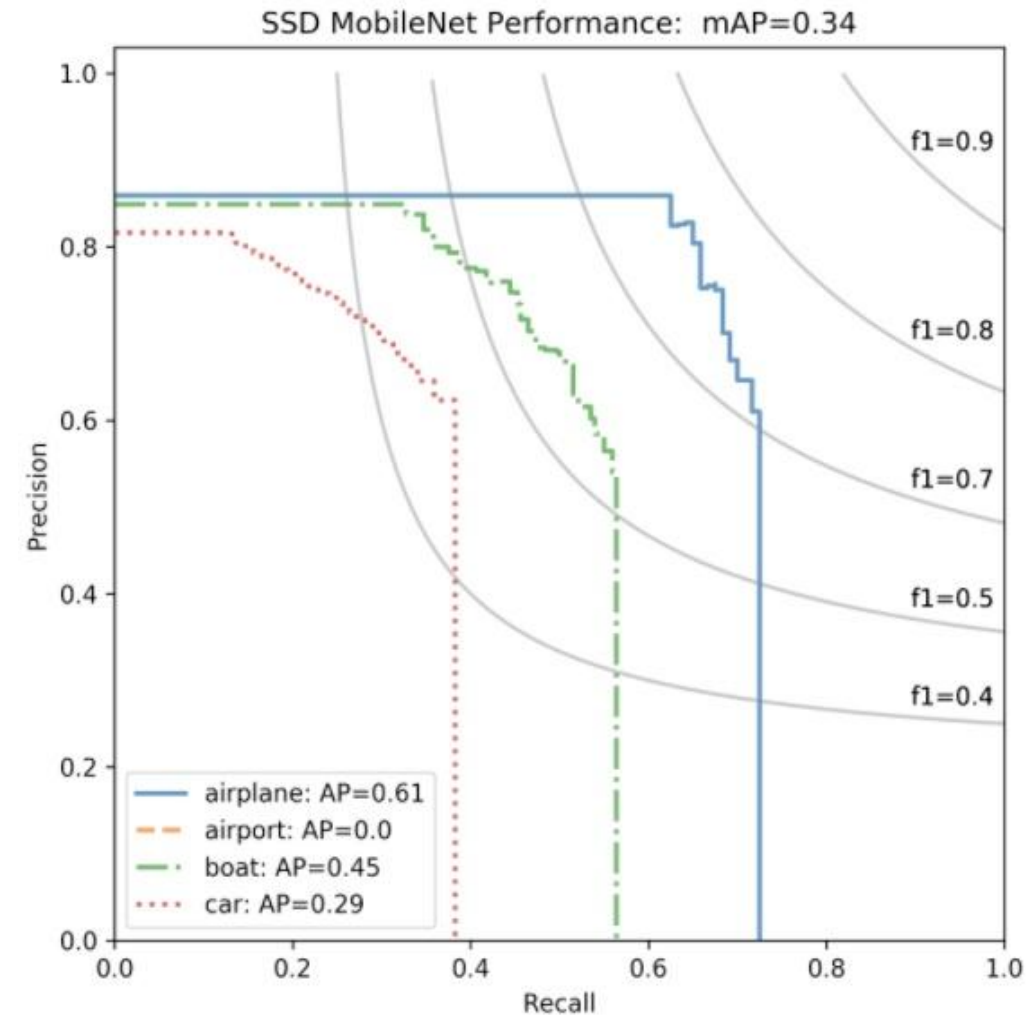
MEAN AVERAGE PRECISION

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

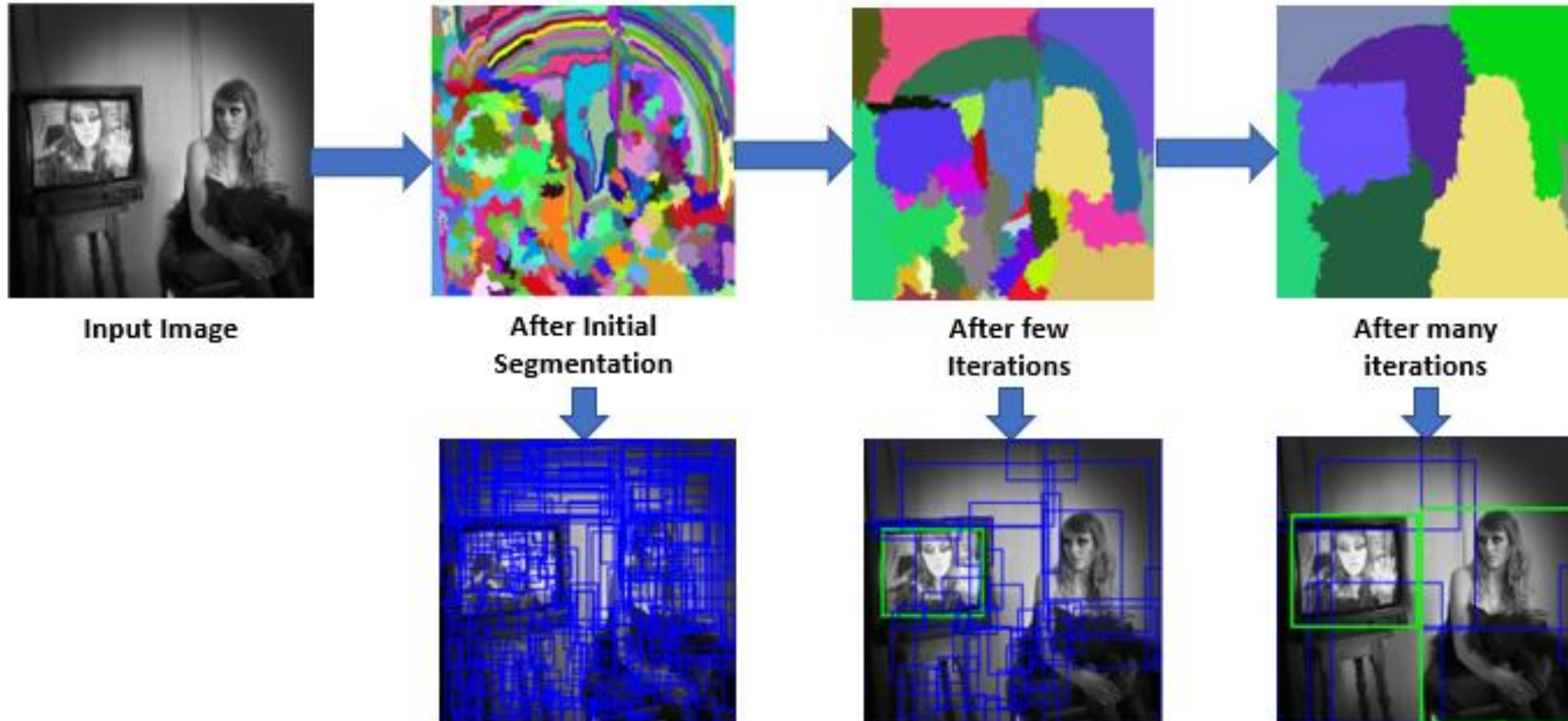
$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$\text{AP}_i \equiv \text{area under curve } i$

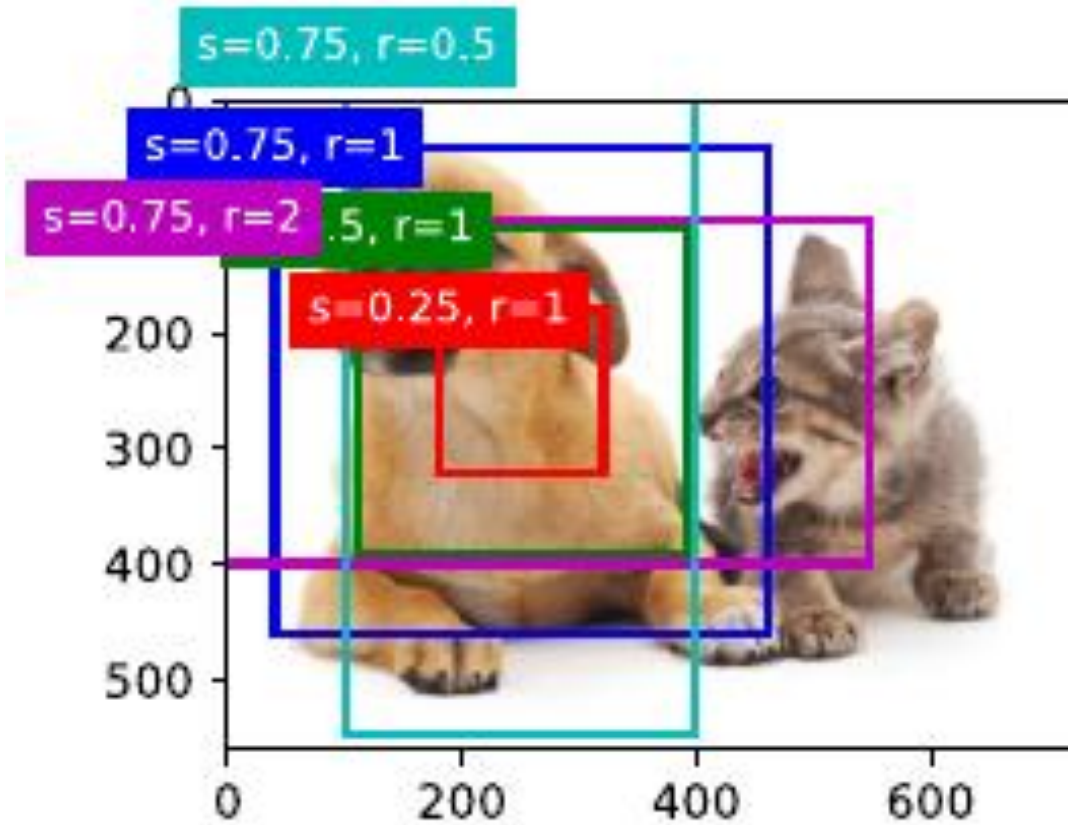
$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i$$



SELECTIVE SEARCH



ANCHOR BOXES



$r > 0$ ratio (width / height)

$s \in (0, 1]$ scale

In practice (s_1, r_1 , scale and ratio references):

$(s_1, r_1), (s_1, r_2), \dots, (s_1, r_m)$

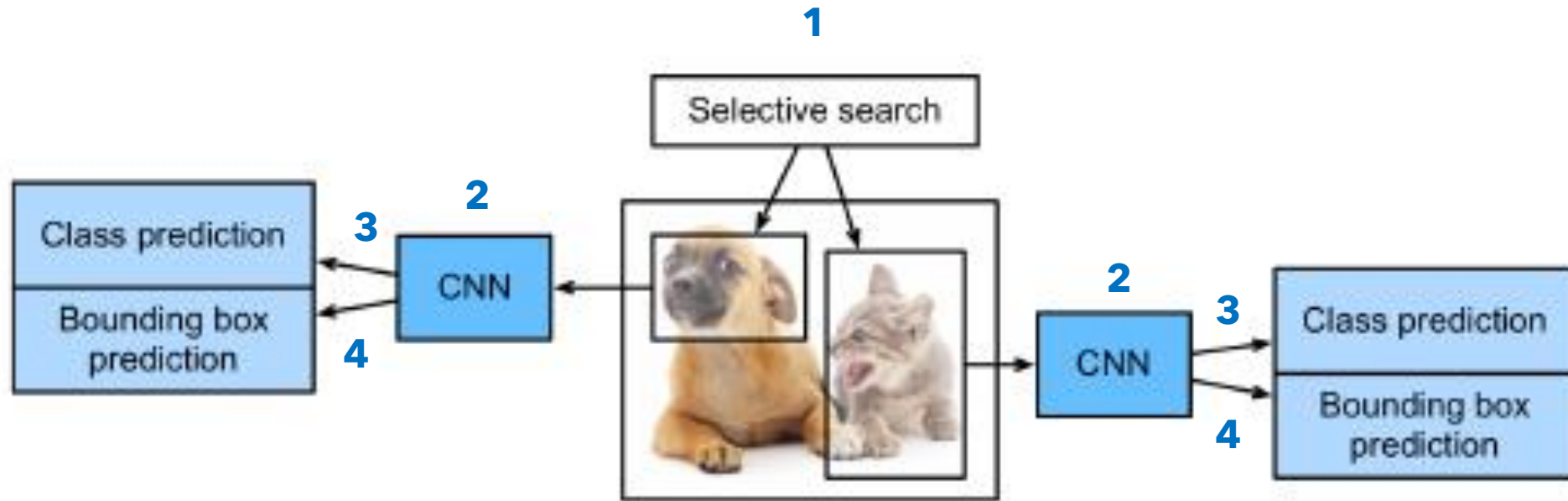
$(s_2, r_1), (s_3, r_1), \dots, (s_n, r_1)$

We pass from whnm to wh(n+m-1)

...

3 . R - C N N S

R-CNNs



5: Translate coordinates to the original image

6: Confidence threshold and NMS



4 . F A S T R - C N N S

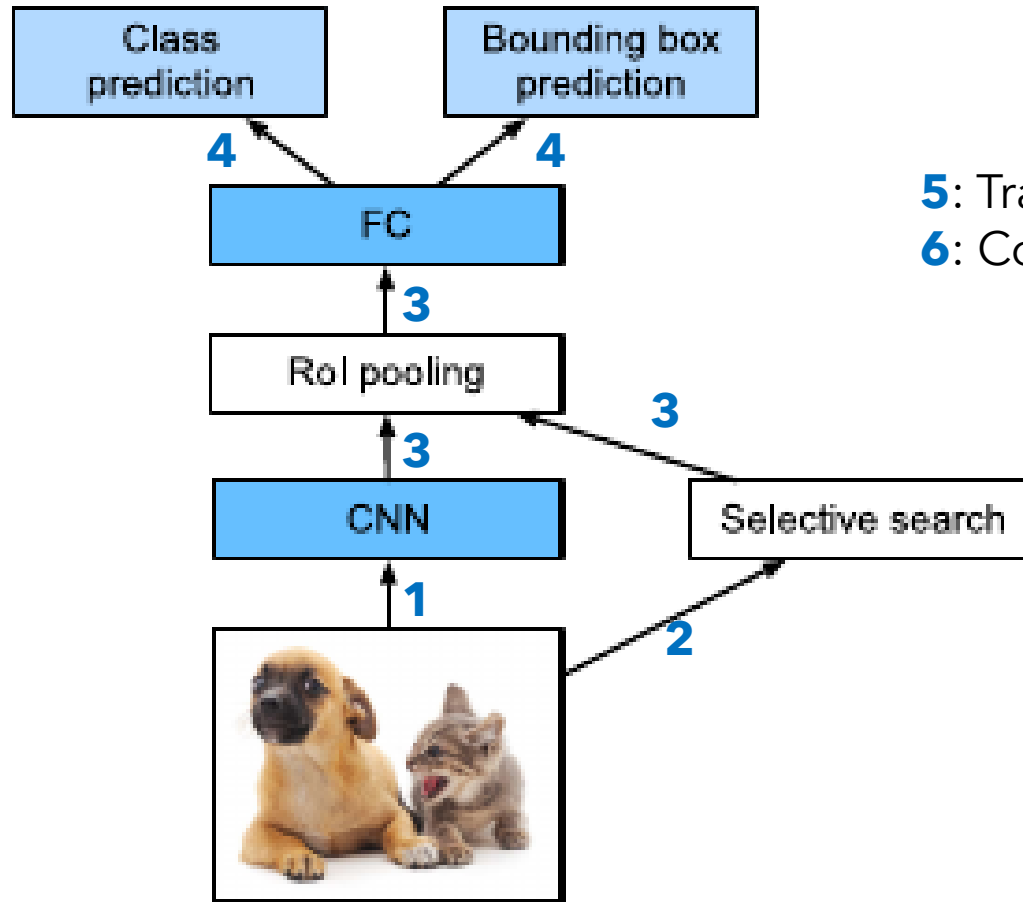
FAST R-CNNs

Problem with R-CNNs: Too many forward passes

Solution: Try to make just one forward pass!



FAST R-CNNs



5: Translate coordinates to the original image

6: Confidence threshold and NMS



FAST R-CNNs

Rol Pooling

Region Proposal

2	1	4
6	5	0
2	3	9

Divide into a Grid

2	1	4	
6	5	0	
2	3	9	

Apply
Aggregation
Function

6	4
3	9



5 . F A S T E R R - C N N S

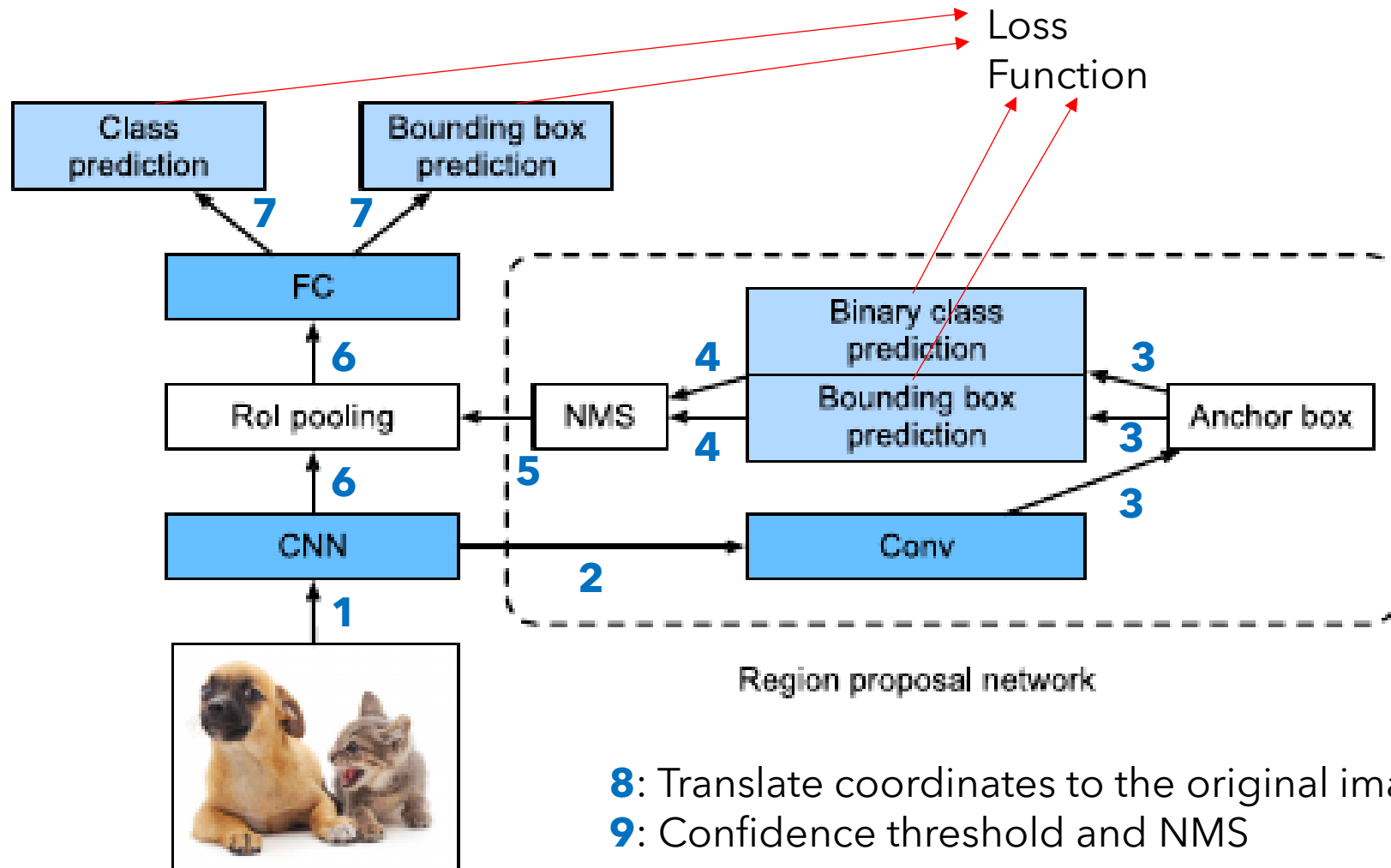
FASTER R-CNNs

Problem with Fast R-CNNs: They need many region proposals to work well

Solution: Try to learn the region proposals!



FASTER R-CNNs



FASTER R-CNNs

	R-CNN	Fast R-CNN	Faster R-CNN
Test time per image	50 seconds	2 seconds	0.2 seconds
Speed-up	1x	25x	250x
mAP (VOC 2007)	66.0%	66.9%	66.9%

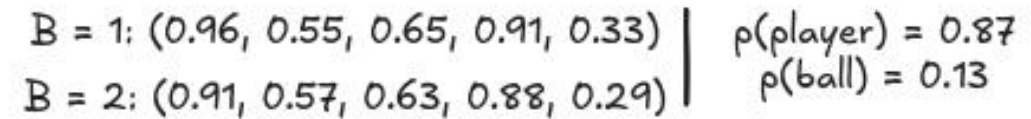


6 . SINGLE STAGE OBJECT DETECTION

YOLO IDEA

1. Divide the image into a grid of $S \times S$
2. Each cell predicts B bounding boxes: $(p(\text{object}), x, y, w, h)$
3. Each cell also predicts the probability of the class of the object: $(p(\text{class}_1), \dots, p(\text{class}_C))$
4. We obtain a tensor of shape $S \times S \times (5B+C)$
5. Choose best bounding boxes and apply NMS





B = 1: (0.87, 0.71, 0.35, 0.16, 0.19)	$p(\text{player}) = 0.19$
B = 2: (0.92, 0.69, 0.32, 0.17, 0.24)	$p(\text{ball}) = 0.81$

$$\begin{array}{l|l} B = 1: (0.03, _, _, _, _) & p(\text{player}) = _ \\ B = 2: (0.01, _, _, _, _) & p(\text{ball}) = _ \end{array}$$

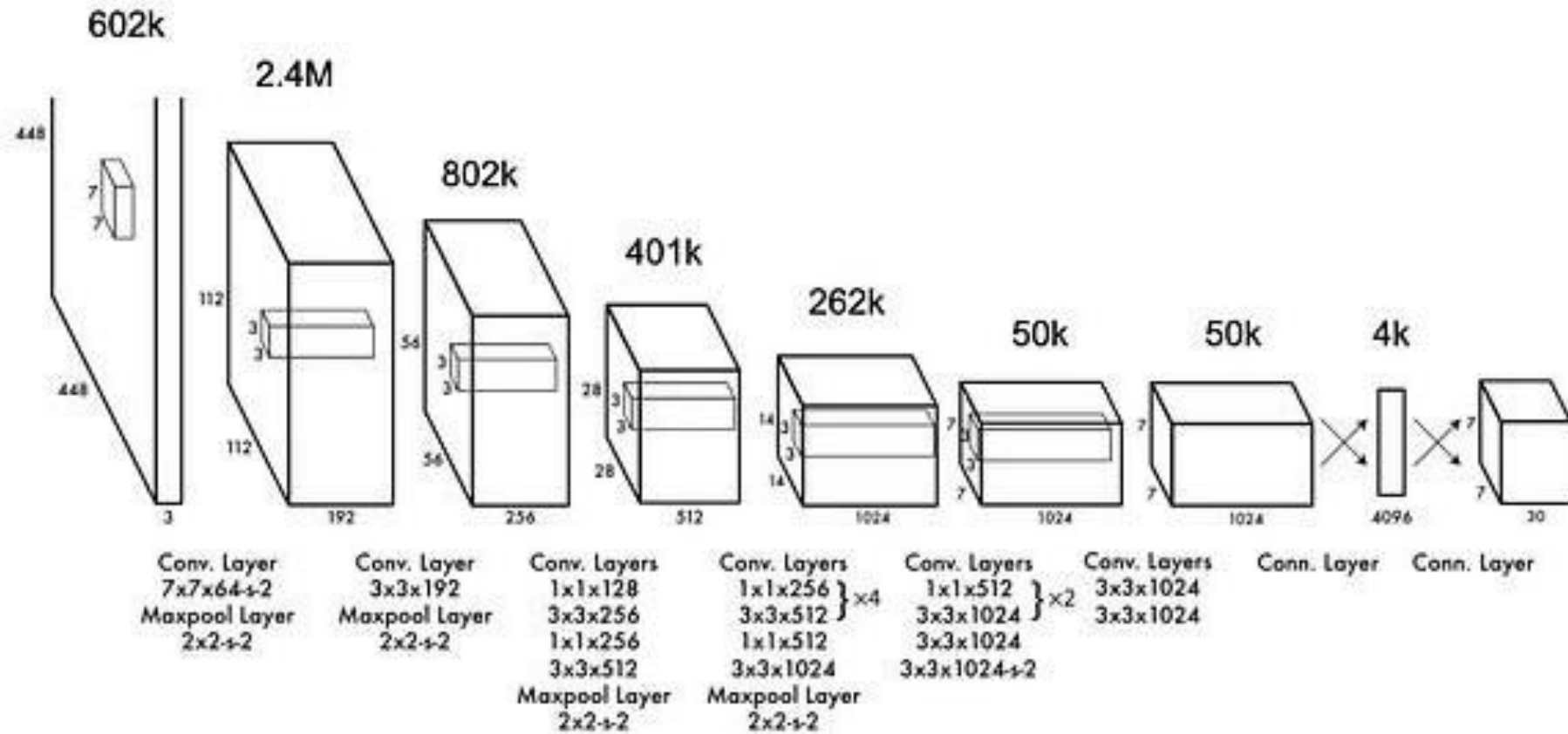
— could be any random number between 0 and 1, because the model has to output always a vector with the same shape, but in practice it does not mean anything, as there is no a predicted object in the first position

YOLO LIMITATIONS

- Only one class per cell
- At most B bounding boxes (of the same class, of course) per cell



YOLO ARCHITECTURE



YOLO LOSS

$$\mathcal{L} = \mathcal{L}_{\text{BB}} + \mathcal{L}_{\text{O}} + \mathcal{L}_{\text{NO}} + \mathcal{L}_{\text{C}}$$

$$\mathcal{L}_{\text{BB}} = \lambda_{\text{coord}} \sum_{i=1}^{S^2} \sum_{j=1}^B \underline{1_{ij}^{\text{obj}}} \left[(x_i - \hat{x}_{ij})^2 + (y_i - \hat{y}_{ij})^2 + \left(\underline{\sqrt{w_i} - \sqrt{\hat{w}_{ij}}} \right)^2 + \left(\underline{\sqrt{h_i} - \sqrt{\hat{h}_{ij}}} \right)^2 \right]$$

$$\mathcal{L}_{\text{O}} = \sum_{i=1}^{S^2} \sum_{j=1}^B \underline{1_{ij}^{\text{obj}}} \left(C_{ij} - \hat{C}_{ij} \right)^2 \quad C_{ij} = \text{IoU} \left(\hat{B}_{ij}, B_i \right) \quad | \quad 1$$

$$\mathcal{L}_{\text{NO}} = \lambda_{\text{noobj}} \sum_{i=1}^{S^2} \sum_{j=1}^B \underline{1_{ij}^{\text{noobj}}} \hat{C}_{ij}^2 = \lambda_{\text{noobj}} \sum_{i=1}^{S^2} \sum_{j=1}^B \left(1 - \underline{1_{ij}^{\text{obj}}} \right) \hat{C}_{ij}^2$$

$$\mathcal{L}_{\text{C}} = \sum_{i=1}^{S^2} \underline{1_i^{\text{obj}}} \sum_{c=1}^C (p_i(c) - \hat{p}_i(c))^2$$



T H A N K S !

